

YOGESH KUMAR

Adm. No. 25MT0635 7986227694
✉ 25mt0635@iitism.ac.in [linkedin.com/in/mecyogi](https://www.linkedin.com/in/mecyogi)



Education

IIT-ISM Dhanbad

Master of technology-Fuel and Energy engineering (CGPA: 8.02 / 10)

Expected May 2027

Dhanbad, Jharkhand

- **Relevant Coursework:** Fuel technology , Biofuel, Alternate energy, Computational fluid dynamics ,Processing of liquid and gaseous fuel,Energy technology

Lovely professional university

Mechanical Enginerring (CGPA: 7.67 / 10)

June 2019

Jalandhar, Punjab

- **Relevant Coursework:** Mechanics of solid, Fluid dynamics, Thermodynamics,Heat and mass transfer, Production engineering, Solid work, Matlab, Industrial engineering,

Hind angle public school

Class XII (Percentage-73.4)

June 2013

Hamirpur, Uttarpradesh

J.D.S.V.M Kanpur

Class-X (CGPA: 8.00 / 10)

May 2011

Kanpur, Uttarpradesh

Experience

Vaidik Eduservices Pvt. Ltd.

Subject matter expert

June 2022 – sep 2024

Noida, Uttar Pradesh

- Domain expertise: Thermodynamics, fuel combustion, hydrogen production and storage (SMR, electrolysis, hydrides), materials characterization (TGA/DTG, XRF).
- Delivery: Data analysis, experimental design, validation protocols, and cross-functional project management with clear metrics and risk controls
- Training, documentation, SOPs, and coaching for non-experts; requirements gathering and solution design with stakeholders and vendors.

Projects

Experimental Investigation of Performance and Emission Characteristics of a CI Engine Using Pine Oil–Diesel Blends

- Investigated the effect of increasing pine-oil concentration in diesel blends on exhaust-gas characteristics, with particular emphasis on CO₂ emissions and overall combustion behavior.
- Evaluated the impact of pine-oil–diesel blending on key engine performance parameters and brake thermal efficiency to determine the influence of fuel composition on engine efficiency
- Identified an optimized 17% pine-oil–diesel blend that provided a balanced combination of emission characteristics, engine efficiency, and fuel-economic benefits for CI-engine applications.

FJH Synthesis of MoO₃–Graphene and V₂O₅–Graphen Hybrid Nanocomposites for High-Performance Supercapacitors

- * Synthesized MoO₃–Graphene and V₂O₅–Graphene hybrid nanocomposites using Flash Joule Heating (FJH) for advanced supercapacitor applications .
- * Characterized the synthesized materials using XRD, Raman, FTIR, SEM/EDS and evaluated their structural, morphological and compositional properties .
- * item Fabricated composite electrodes and performed CV, GCD and EIS analyses to determine specific capacitance, energy/power density, coulombic efficiency and cycling stability .
- * Investigated the synergistic contribution of graphene conductivity and metal-oxide pseudocapacitance to enhance electrochemical charge-storage performance .

Technical Skills

Languages: C, C++, Latex, MS-Excel, MS-word

Software: Matlab, ANSYS-FLUENT, Origin, COMSOL,

Technical skill: RAMAN SPECTROSCOPY, XRD, TEM, SEM, XPS, BET, FJH

Concepts: Coal and mineral beneficiation, Computational Fluid dynamics, Proximate analysis, Ultimate analysis, Flame Theory, TGA, XRF, Analysis of coal and biomass, Bio-fuel production

Achievements

* JEE - Mains - 2014

* Gate -2025

Social Engagements

Conferences: CRISP-Critical Metals: Recycling, Innovation, Separation and Processing , IIT Dhanbad

Volunteer: At Coke studio(Tech fest) - LPU ,CRISP 2025-IIT Dhanbad

Sports-Engagements: Badminton, cricket ,Football.